MS&E 342 - Spring 2026
Stochastic Systems and Learning Theory with Applications in Finance
Course Syllabus

04/20/2026

General Information

- **Class Meeting Times:** Tuesdays and Thursdays 3:00 - 4:20 PM
- **Location:** Lane History Corner (Building 200) Room 34
- **Instructor**: Renyuan Xu
  - Email: renyuanxu@stanford.edu
  - Regular office hours: **9-10am Thursday**, in person at Huang 355
  - Additional office hours (by appointment only): I am available to hold additional office hours on some Thursdays from 4:30–6:00pm. Please email me in advance if you would like to schedule a meeting.
- **Target audience:** PhD students

Course Contents

This course studies stochastic systems and learning theory with applications in finance. The primary goal is to provide a unified mathematical framework connecting stochastic analysis, stochastic control, and modern machine learning methods. We begin with a review of stochastic calculus and introduce the stochastic foundations of generative diffusion models. The course then develops continuous-time stochastic control theory, including Hamilton–Jacobi–Bellman equations, the maximum principle, and optimal stopping, together with their applications in financial decision-making. Building on these foundations, we explore reinforcement learning in continuous time and its connections to modern approaches for fine-tuning generative AI models. Finally, the course introduces large-scale stochastic games, including mean field games and mean field control, which provide tractable models for systems with many interacting agents. Throughout the course, we emphasize both rigorous mathematical formulations and their relevance to modern problems in finance and machine learning.

Prerequisites

Students are expected to have background in:

- Stochastic calculus
- Functional analysis
- Ordinary and partial differential equations
- Machine learning and statistics

Course Materials

The primary course materials are the lecture notes and slides. The recommended tutorials/textbooks for this course are

- Tang, Wenpin, and Hanyang Zhao. "Score-based diffusion models via stochastic differential equations." Statistic Surveys 19 (2025): 28-64.
- Pham, Huyên. Continuous-time stochastic control and optimization with financial applications. Vol. 61. Springer Science & Business Media, 2009.
- Lacker, Daniel. "Mean field games and interacting particle systems." preprint (2018).
- Uehara, Masatoshi, et al. "Understanding reinforcement learning-based fine-tuning of diffusion models: A tutorial and review." arXiv preprint arXiv:2407.13734 (2024).
- Chi, Yuejie, Yuxin Chen, and Yuting Wei. "Statistical and algorithmic foundations of reinforcement learning." Tutorials in Operations Research: Advances in Analytics and Operations Research: Improving Decisions to Secure the Future. INFORMS, 2025. 104-144.
- Hambly, Ben, Renyuan Xu, and Huining Yang. "Recent advances in reinforcement learning in finance." Mathematical Finance 33.3 (2023): 437-503.

Course Website

We will manage course materials through the Canvas sites:

`https://canvas.stanford.edu`

Lecture slides are posted on that site, and messages to the class will be delivered through that site. Students are required to enroll on this website. Based on our experience some students prefer the possibility to ask course related questions in an anonymous discussion forum. Unfortunately, Canvas does not offer this option. Therefore we have created an Ed Discussion module which you can access from the Canvas site. Ed Discussion is a forum that allows students and teaching staff to discuss the course effectively. Ed Discussion is the preferred method of communication for course-related questions.

Homework and project assignments should be submitted on Gradescope:

`https://www.gradescope.com/courses/1284208`

Gradescope offers better functionality for group work and for feedback on your submitted assignments than Canvas. We will try to add you to Gradescope but you can also sign yourself up with the entry code **RV24XN**. Your final scores will also be posted on the Canvas website.

Course Units

The workload of this course corresponds to a 3 unit master level course. You can choose a letter grade or pass/fail. You will pass this course if you have a letter grade of C- or better.

Requirements

Grading The weights for grading are

- Lecture attendance: 30%
- In-class paper presentation (two co-presenters for each paper): 30%
- Choose one from the following (40%):
  - Two homework assignments (completed individually)
  - Final project (up to two team members)

If you are in between two grades there is the possibility to improve your grade through good participation in class, office hours and/or Ed Discussion forums.

Homework

- 2 homework assignments will be uploaded on Thursdays.
- Homework should be completed and submitted individually.
- Please submit your homework on Gradescrope.
- Homework assignments are usually due one week later on Sunday at 11pm. The submitted homework can be scanned versions of handwritten solutions or typed solutions (Word, Latex, etc.). Please submit only one pdf file and try to keep a moderate file size.
- Homework will be graded on a 10 point scale.
- You may use financial calculators, spreadsheets, or other computational tools to solve homework problems. However, all answers must be clearly justified.

Lecture attendance

- Attendance is required for at least 12 lectures, which accounts for30% of the final grade.

In-class paper presentation

- Students will present in groups of two. Each presentation will be 30 minutes long, followed by a 5-minute Q&A session.
- Recommended papers are listed at the end of the syllabus. Students may also choose another paper, subject to instructor approval and provided that it is relevant to the course material.
- The presentation topic should be connected to the course part currently being taught.
- Please confirm your selected paper with the instructor before preparing the presentation.

Sign-up the presentation slot here:
https://docs.google.com/document/d/1WuMVhBVxqhWI6Dw13JIFcCmiaidMVXQKIVd8cfx2VSw/edit?usp=sharing

Final project

- Final project can be completed either individually or in teams, with up to two students per team.
- The topic of the project can be broadly related to the contents covered in the course.
- Please confirm the topic and scope of your project with the instructor before working on it.
- You are expected to submit a report (up to 10 pages) for your final project, which is due on **June 12th**.

Honor Code

We expect the students to create an environment in which honor code violation is not to be tolerated. For the exam, this means all work must be done individually. For homework, students may consult with other students, but must write up solutions independently based on their own understanding.

Contents Breakdowns and Timetable

The course is divided into three parts:

- Part I: stochastic calculus and modern applications in generative diffusion models.
- Part II: stochastic control, data-driven decision-making, and financial applications.
- Part III: large-scale stochastic games.

The detailed timetables for lectures and homework are provided below.

Lecture content schedule (expected):

Lec.	Date	Content	Remarks
Part I: Stochastic calculus and modern applications			
1	3/31	Review of stochastic calculus	
2	4/2	Stochastic foundations of generative diffusion models	
3	4/7	Stochastic foundations of generative diffusion models	
4	4/9	Generative diffusion models in finance (45 mins)	Fei Sha's talk at ISL Colloquium at 4pm on diffusion models
5	4/14	Student paper presentations (2 groups)	
Part II: Stochastic control, finance, and ML			
6	4/16	Introduction to stochastic control in continuous time	
7	4/21	HJB equations (and financial applications)	(online; recorded)
8	4/23	Maximum principle (and financial applications)	
9	4/28	Optimal stopping and sequential hypothesis testing	
10	4/30	Student paper presentations (2 groups)	
11	5/5	Continuous-time RL	
12	5/7	Continuous-time RL in finance	
13	5/12	RLHF (fine-tuning generative AI models)	
14	5/14	Student paper presentation (1 group)	Molei Tao's talk at ISL Colloquium at 4pm on diffusion models
15	5/19	Student paper presentations (2 groups)	
Part III: Large-scale stochastic games			
16	5/21	Continuous-time stochastic games	
17	5/26	Mean field games and financial applications	
18	5/28	Mean field controls and financial applications	(online; recorded)
19	6/2	Student paper presentations (2 groups)	(online; recorded)

Homework schedule (expected):

- Homework 1:
  - Release: 4/16 Thursday
  - Due: 5/1 Sunday
- Homework 2:
  - Release: 5/21 Thursday
  - Due: 5/31 Sunday

Recommended papers for presentation

Part I: Diffusion models from stochastic analysis aspects

- Guo, Zhengyi, Wenpin Tang, and Renyuan Xu. "Conditional Diffusion Guidance under Hard Constraint: A Stochastic Analysis Approach." arXiv preprint arXiv:2602.05533 (2026).
- Fu, Hengyu, et al. "Diffusion transformer captures spatial-temporal dependencies: A theory for gaussian process data." arXiv preprint arXiv:2407.16134 (2024).
- Huang, Zhihan, Yuting Wei, and Yuxin Chen. "Denoising diffusion probabilistic models are optimally adaptive to unknown low dimensionality." Mathematics of Operations Research (2026).
- Albergo, Michael, Nicholas M. Boffi, and Eric Vanden-Eijnden. "Stochastic interpolants: A unifying framework for flows and diffusions." Journal of Machine Learning Research 26.209 (2025): 1-80.
- Gao, Xuefeng, Jiale Zha, and Xun Yu Zhou. "Data-driven generative simulation of SDEs using diffusion models." arXiv preprint arXiv:2509.08731 (2025).

Part II: Stochastic control, financial decision-making, and ML

Optimal stopping:

- Karatzas, Ioannis. "Gittins indices in the dynamic allocation problem for diffusion processes." The Annals of Probability 12.1 (1984): 173-192.
- Ekström, Erik, Ioannis Karatzas, and Juozas Vaicenavicius. "Bayesian sequential least-squares estimation for the drift of a Wiener process." Stochastic Processes and their Applications 145 (2022): 335-352.

Financial decision-making:

- Blanchet, Jose, Lin Chen, and Xun Yu Zhou. "Distributionally robust mean-variance portfolio selection with Wasserstein distances." Management science 68.9 (2022): 6382-6410.
- Brokmann, Xavier, et al. "Tackling nonlinear price impact with linear strategies." Mathematical Finance 35.2 (2025): 422-440
- Aghapour, Ahmad, Erhan Bayraktar, and Fengyi Yuan. "Solving dynamic portfolio selection problems via score-based diffusion models." arXiv preprint arXiv:2507.09916 (2025).
- Ning, Brian, Franco Ho Ting Lin, and Sebastian Jaimungal. "Double deep q-learning for optimal execution." Applied Mathematical Finance 28.4 (2021): 361-380.

Continuous-time RL and fine-tuning:

- Basei, Matteo, et al. "Logarithmic regret for episodic continuous-time linear-quadratic reinforcement learning over a finite-time horizon." Journal of Machine Learning Research 23.178 (2022): 1-34.

- Jia, Yanwei, and Xun Yu Zhou. "Policy evaluation and temporal-difference learning in continuous time and space: A martingale approach." Journal of Machine Learning Research 23.154 (2022): 1-55
- Cheng, Ziheng, Xin Guo, and Yufei Zhang. "Bridging Discrete and Continuous RL: Stable Deterministic Policy Gradient with Martingale Characterization." arXiv preprint arXiv:2509.23711 (2025).
- Domingo-Enrich, Carles, et al. "Adjoint matching: Fine-tuning flow and diffusion generative models with memoryless stochastic optimal control." arXiv preprint arXiv:2409.08861 (2024).
- Garg, Jhanvi, Xianyang Zhang, and Quan Zhou. "Soft-constrained Schrödinger Bridge: a stochastic control approach." International Conference on Artificial Intelligence and Statistics. PMLR, 2024.

Part III: Large-scale stochastic games

- Iyer, Krishnamurthy, Ramesh Johari, and Mukund Sundararajan. "Mean field equilibria of dynamic auctions with learning." Management Science 60.12 (2014): 2949-2970.
- Calvano, Emilio, et al. "Artificial intelligence, algorithmic pricing, and collusion." American Economic Review 110.10 (2020): 3267-3297.
- Cardaliaguet, Pierre, and Charles-Albert Lehalle. "Mean field game of controls and an application to trade crowding." Mathematics and Financial Economics 12.3 (2018): 335-363.
- Han, Jiequn, and Qianxiao Li. "A mean-field optimal control formulation of deep learning." Research in the Mathematical Sciences 6.1 (2019): 1-41.
- Qu, Guannan, Adam Wierman, and Na Li. "Scalable reinforcement learning for multiagent networked systems." Operations Research 70.6 (2022): 3601-3628.